

Course: MMDS

Lab 03

MapReduce

Fill answers of the questions below in the given tables.

*Your screenshots must **contain commands** for required operations.*

Question 1:

Given a tsv file [WHO-COVID-19-20210601-213841.tsv](#) which is corresponding to the [WHO Coronavirus \(COVID-19\) Dashboard](#).

Students are required to create a folder, named **lab03**, in HDFS and then copy the tsv to **lab03/input/**

Take a screenshot to show the content of **lab03/input/** in HDFS

```
doanquochuy@Huys-MacBook ~ % docker exec -it namenode bash
root@a3ba835a2db5:/# hdfs dfs -mkdir -p /user/doanquochuy/lab03/input
root@a3ba835a2db5:/# exit
exit
doanquochuy@Huys-MacBook ~ % docker cp /Users/doanquochuy/hadoop-ueh/WHO-COVID-19-20210601-213841.tsv namenode:/tmp/
Successfully copied 30.7kB to namenode:/tmp/
doanquochuy@Huys-MacBook ~ % docker exec -it namenode bash
root@a3ba835a2db5:/# hdfs dfs -put /tmp/WHO-COVID-19-20210601-213841.tsv /user/doanquochuy/lab03/input
2026-04-20 12:59:05,075 INFO sasl.SaslDataTransferClient: SASL encryption trust check: localHostTrusted = false, remoteHostTrusted = false
root@a3ba835a2db5:/# hdfs dfs -ls /user/doanquochuy/lab03/input/
Found 1 items
-rw-r--r-- 1 root supergroup 28907 2026-04-20 12:59 /user/doanquochuy/lab03/input/WHO-COVID-19-20210601-213841.tsv
root@a3ba835a2db5:/#
```

Question 2:

Create one and only one java file, named **ASEANCaseCount.java**, to run a MapReduce job that counts the number of cumulative total cases among ASEAN countries (*South-East Asia Region in the given data table*).

The output of the MapReduce job is located in **lab03/output-java/**.

Submit the source code file following the instructions in Submission Notice.

Question 3 (optional):

Create a pair of Python files, named **ASEANDeathCountMapper.py** and **ASEANDeathCountReducer.py**, to run a MapReduce job that counts the number of cumulative total deaths among ASEAN countries (*South-East Asia Region in the given data table*). The output of the MapReduce job is located in **lab03/output-python/**. Submit the source code files following the instructions in Submission Notice.

Submission Notice

- Export your answer file as pdf
- Rename the pdf following the format:
lab03_<studentnumber>_<full name>.pdf
E.g. lab03_123456_NguyenThanhAn.pdf
If you have not been assigned a student number yet, then use 123456 instead.
- Create a folder with the name as **lab03_<student number>_<full name>**, which contains
 - **lab03_<studentnumber>_HoTen.pdf** → your answer
 - java/ | → Java sourcecode folder
| **ASEANCaseCount.java**
 - python/ | → Python source code folder
| **ASEANDeathCountMapper.py**
| **ASEANDeathCountReducer.py**
- Compress the folder into a zip file and finally submit to the given form.
- Careless mistakes in filename, format, question order, etc. are not accepted (0 pts).